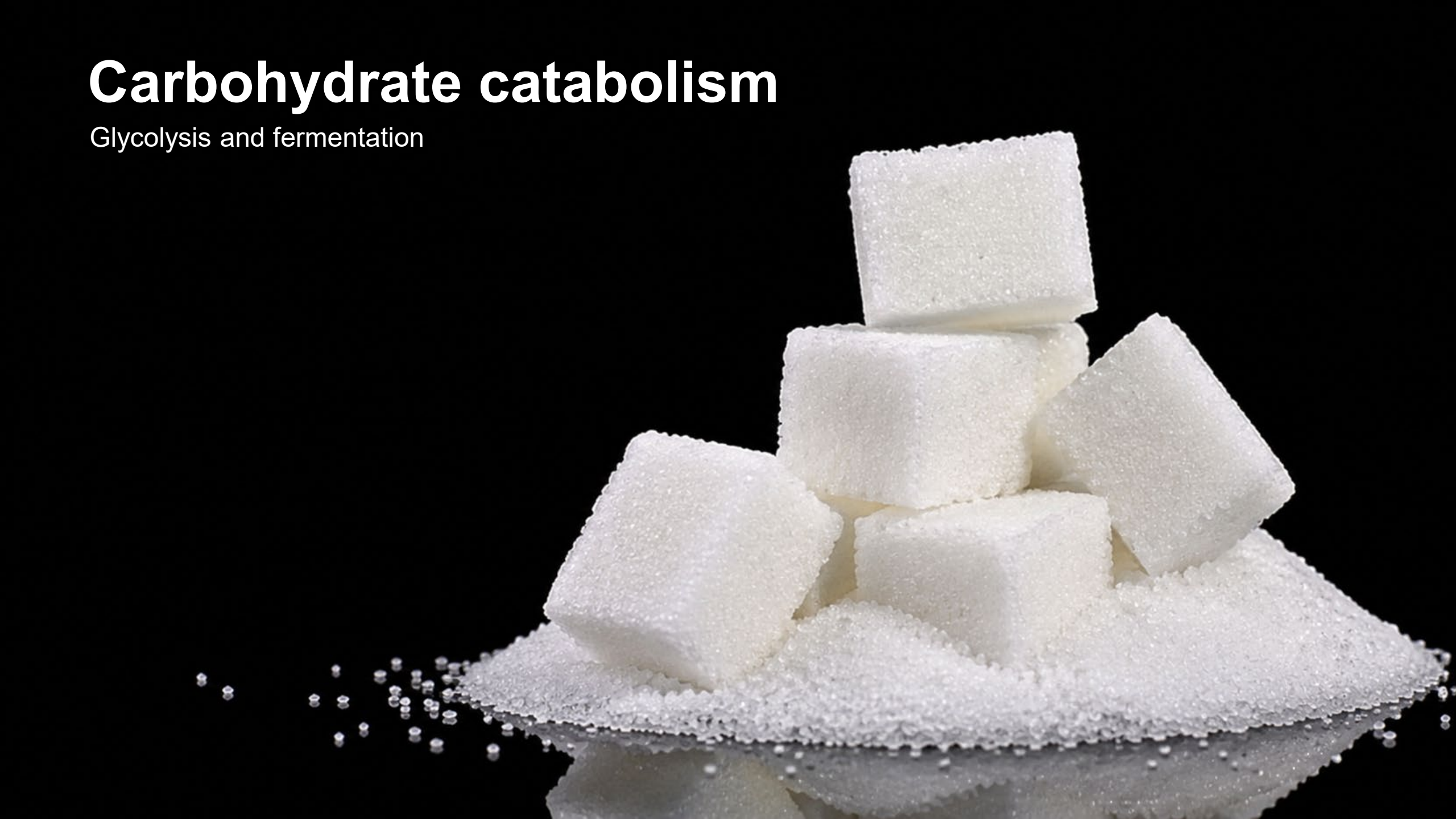
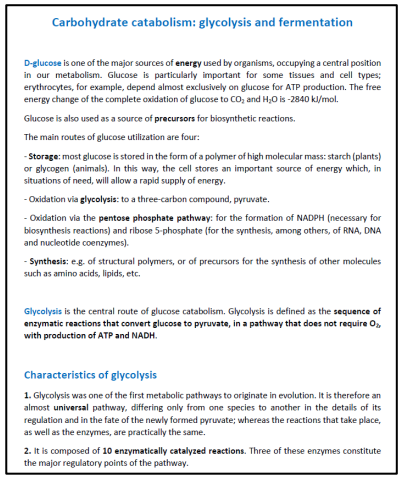
Carbohydrate catabolism

Glycolysis and fermentation



Supportive material



Carbohydrate catabolism: glycolysis and fermentation

D-glucose is one of the major sources of **energy** used by organisms, occupying a central position in our metabolism. Glucose is particularly important for some tissues and cell types; erythrocytes, for example, depend almost exclusively on glucose for ATP production. The free energy change of the complete oxidation of glucose to CO_2 and H_2O is -2840 kJ/mol.

Glucose is also used as a source of **precursors** for biosynthetic reactions.

The main routes of glucose utilization are four:

- **Storage**: most glucose is stored in the form of a polymer of high molecular mass: starch (plants) or glycogen (animals). In this way, the cell stores an important source of energy which, in situations of need, will allow a rapid supply of energy.

- Oxidation via **glycolysis**: to a three-carbon compound, pyruvate.

- Oxidation via the **pentose phosphate pathway**: for the formation of NADPH (necessary for biosynthesis reactions) and ribose 5-phosphate (for the synthesis, among others, of RNA, DNA and nucleotide coenzymes).

- **Synthesis:** e.g. of structural polymers, or of precursors for the synthesis of other molecules such as amino acids, lipids, etc.

Glycolysis is the central route of glucose catabolism. Glycolysis is defined as the **sequence of enzymatic reactions that convert glucose to pyruvate, in a pathway that does not require O_2, with production of ATP and NADH.**

Characteristics of glycolysis

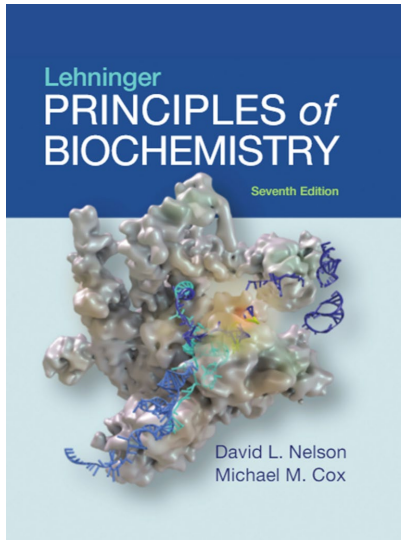
1. Glycolysis was one of the first metabolic pathways to originate in evolution. It is therefore an almost **universal** pathway, differing only from one species to another in the details of its regulation and in the fate of the newly formed pyruvate; whereas the reactions that take place, as well as the enzymes, are practically the same.

2. It is composed of **10 enzymatically catalyzed reactions**. Three of these enzymes constitute the major regulatory points of the pathway.

Additional study material

ADI document:

Carbohydrate catabolism



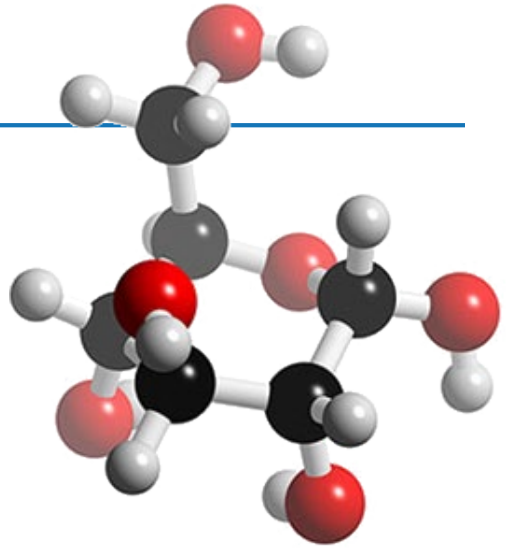
Reading material

Lehninger's PoB Chapter 14: 14.1., 14.2. and 14.3.

Learning goals

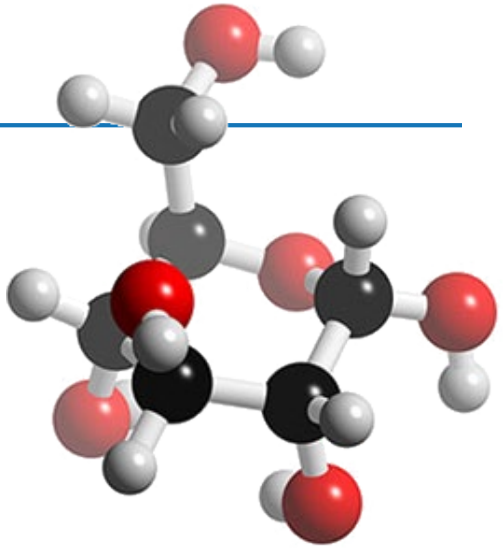
- Steps for the generation of pyruvate from glucose
- Hexokinase vs. glucokinase
- Regulation of glycolysis
- Fermentations
- Overview of the entry of other sugars into glycolysis

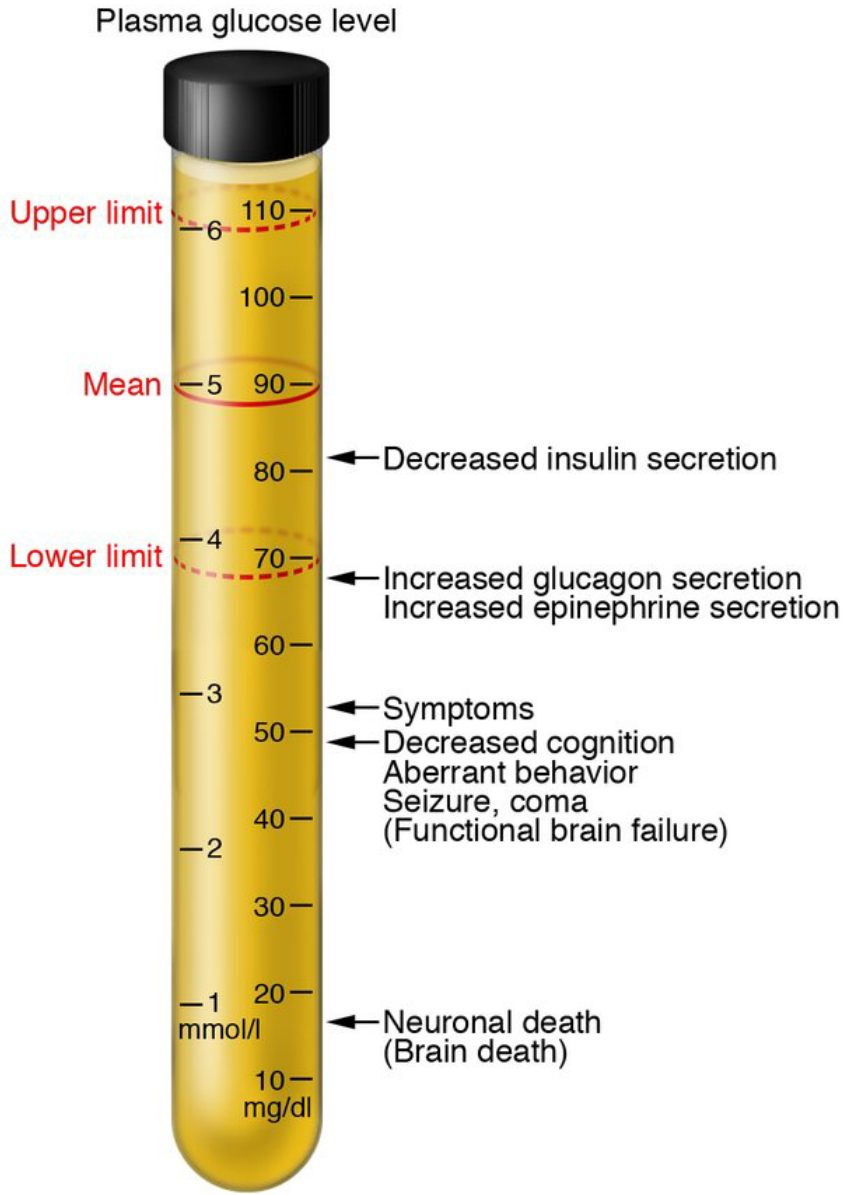
The importance of glucose



- Glucose is an excellent source of energy
  - Yields a good amount of energy upon oxidation
  - Can be efficiently stored in a polymeric form
- Glucose is a versatile precursor that can be used to build the carbon skeletons of:
  - Amino acids
  - Membrane lipids
  - Nucleotides
  - Other carbohydrates, cofactors …

The importance of glucose





Four major pathways of glucose utilization

- Storage
  Can be stored in a polymeric form (glycogen, starch)
- Oxidation via glycolysis
  Generates energy via oxidation of glucose
  Short-term energy needs
- Oxidation via pentose phosphate pathway
  Generates NADPH and pentoses via oxidation of glucose
  For the biosynthesis of lipids and nucleotides
- Synthesis

Glycolysis

- Sequence of enzyme-catalyzed reactions by which glucose is converted into pyruvate

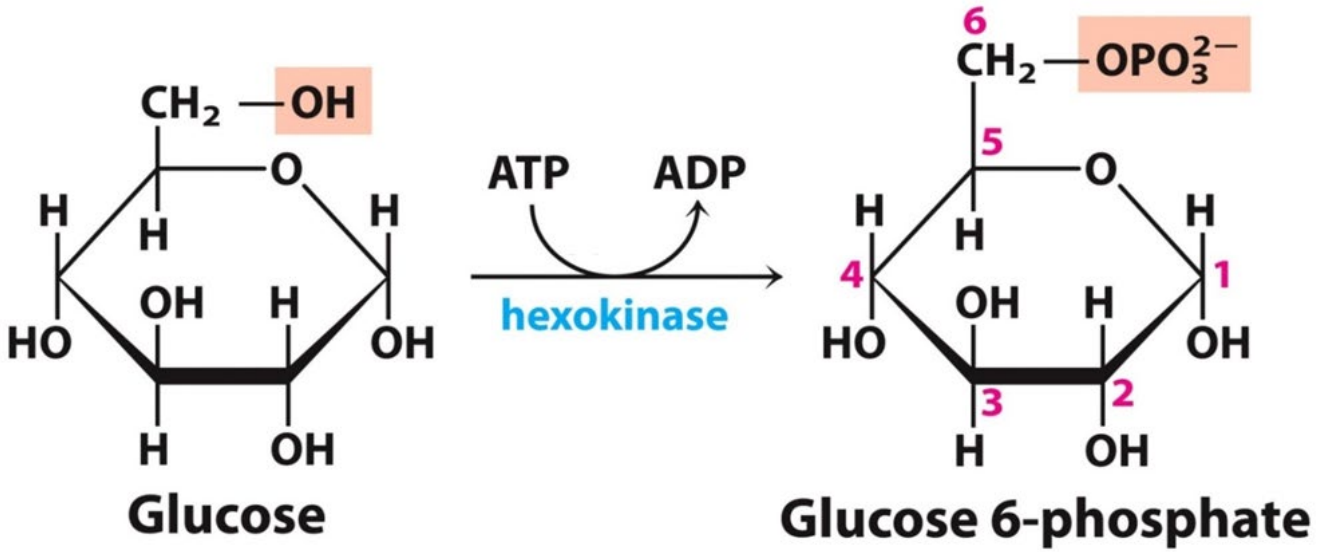
$$\text{Glucose } (C_6H_{12}O_6) \rightarrow 2 \text{ pyruvate } (C_3H_4O_3)$$

- Some of the oxidation-free energy is captured by the synthesis of ATP and NADH

Glycolysis: characteristics

- Universal
- Ten steps, three of them tightly regulated
- Both catabolic and anabolic
- Cytoplasmic
- Irreversible
- It does not require oxygen
- All intermediate compounds are phosphorylated

Step 1. Glucose phosphorylation



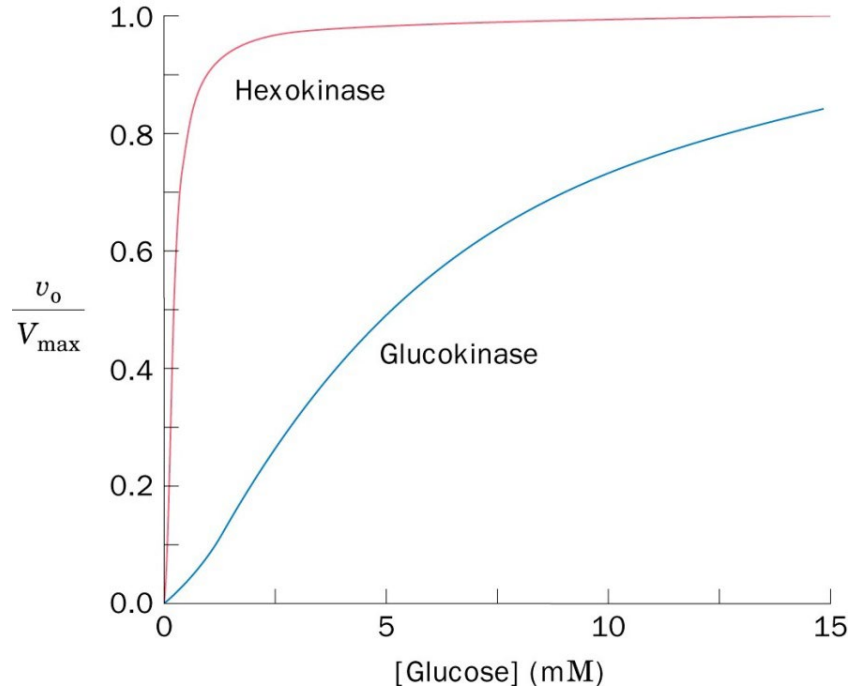
Lehninger 7th ed.

$\Delta G'° = -16.7$ kJ/mol

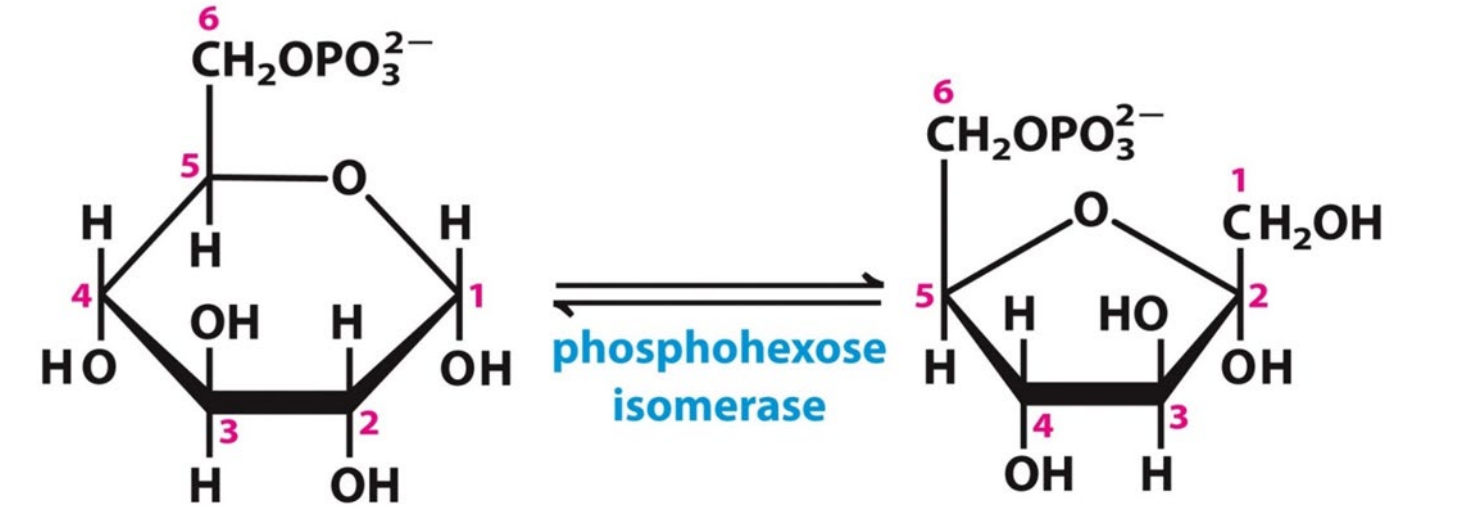
- **Aim**:
  - faTraps glucose inside the cell
  - Increases the free-energy content of glucose by phosphorylation
- Highly thermodynamically vorable/irreversible
- This step uses the energy of ATP

Hexokinase isoforms

- The human genome contains four different hexokinases (I to IV)
- HK I is expressed in all tissues
- HK IV (glucokinase) is expressed in liver and pancreas
  - Has higher K_m, so responses to higher [glucose]
  - Has higher specificity for glucose
  - Not inhibited by glucose 6-phosphate



Step 2. Hexose phosphate interconversion



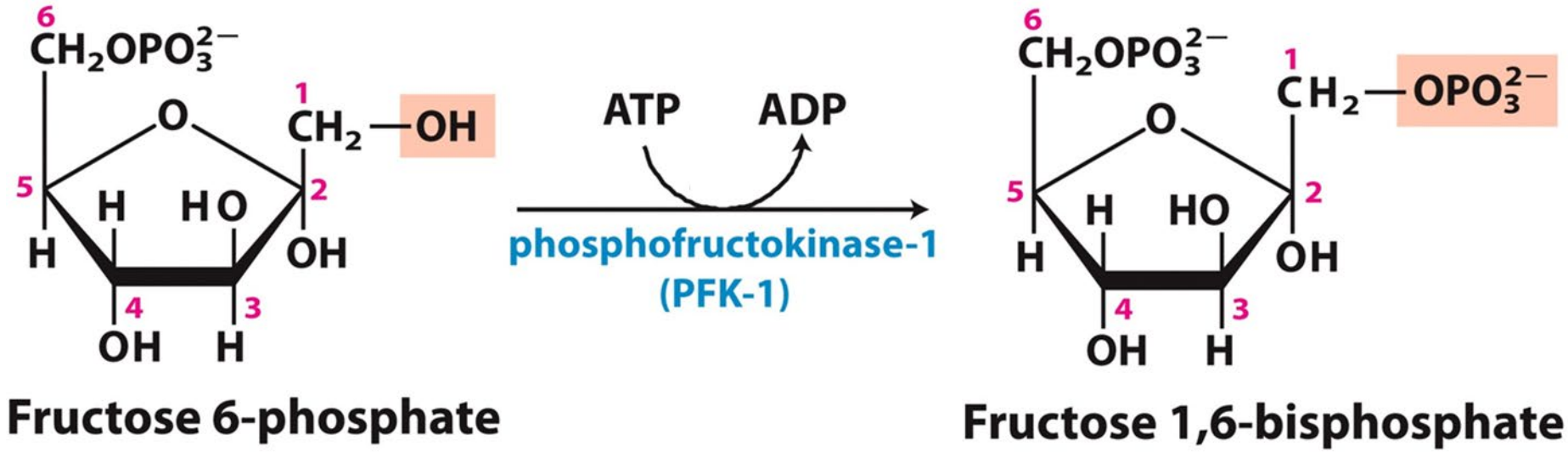
Glucose 6-phosphate **Fructose 6-phosphate**

Lehninger 7th ed.

$\Delta G'^{\circ} = 1.7$ kJ/mol

- **Aim**:
  - Allows subsequent symmetrical cleavage
- Isomerization
- Reversible

Step 3. Phosphorylation of fructose 6-phosphate

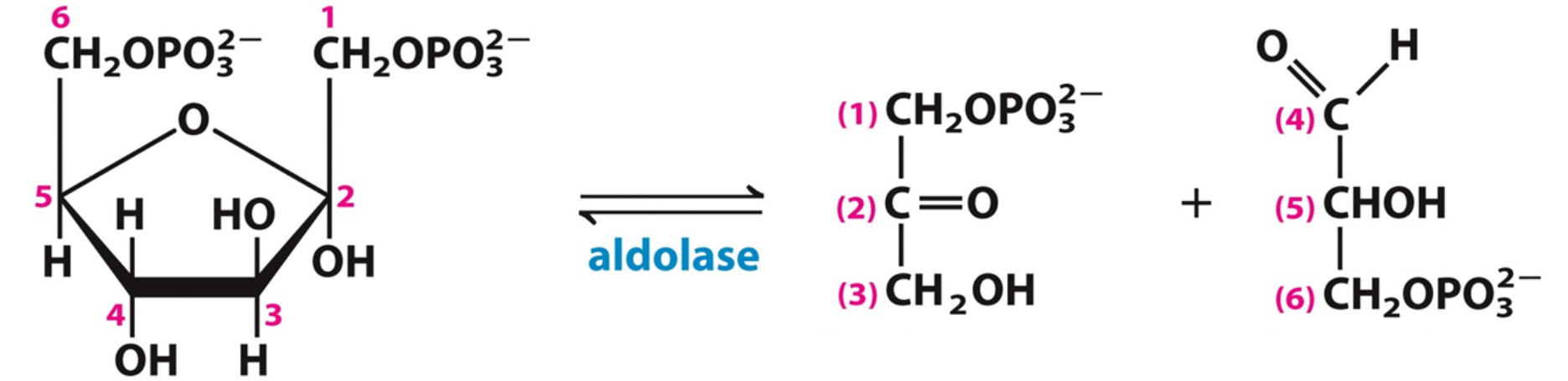


Lehninger 7th ed.

$\Delta G'^{\circ} = -14.2$ kJ/mol

- **Aim**:
  - Further activates the intermediate
  - Allows for symmetrical cleavage
- Highly thermodynamically favorable/irreversible (it is the commitment step in glycolysis)
- This step uses the energy of ATP

Step 4. Cleavage of fructose 1,6-bisphosphate



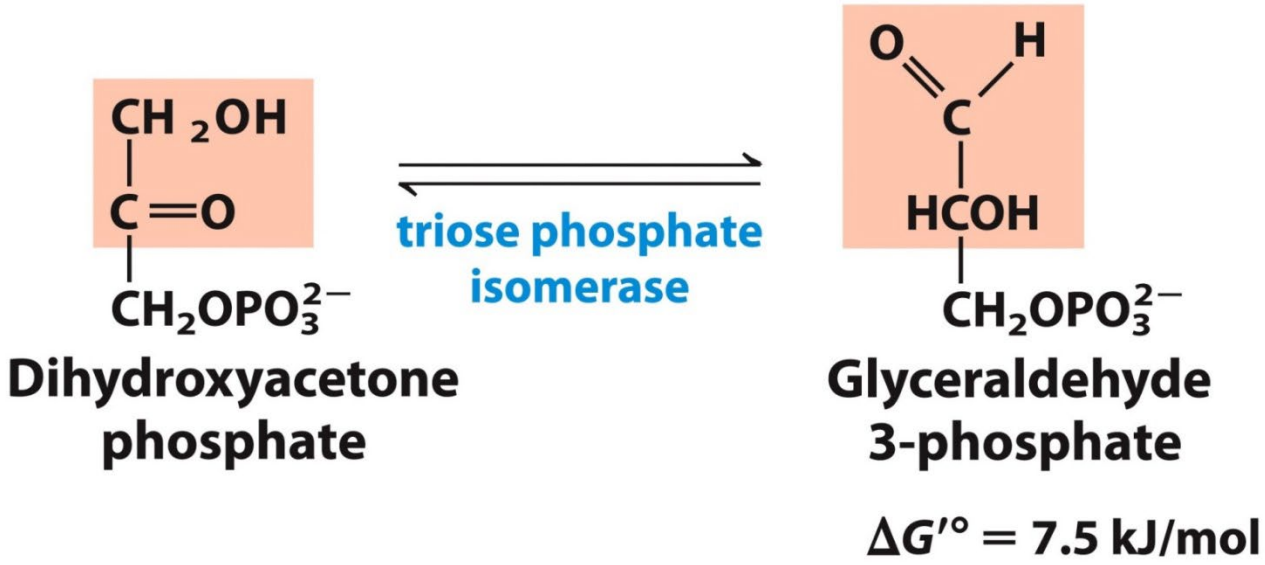
Fructose 1,6-bisphosphate | **Dihydroxyacetone phosphate** | **Glyceraldehyde 3-phosphate**

Lehninger 7th ed.

$\Delta G'^{\circ} = 23.8$ kJ/mol

- **Aim**:
  - Generates three-carbon molecules
- Reversible (because the cytosolic concentration of glyceraldehyde 3-phosphate is very low)

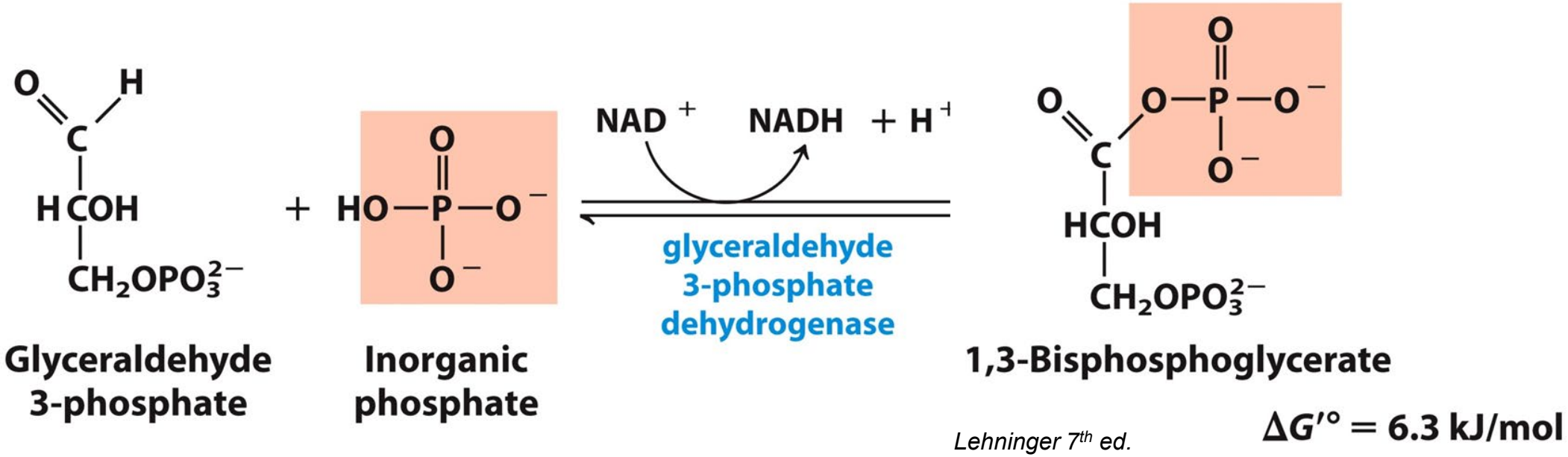
Step 5. Triose phosphate interconversion



Lehninger 7th ed.

- **Aim**:
  - Allows glycolysis to proceed by a unique pathway (from two GAP molecules)
- Isomerization
- Reversible (intracellular concentration of GAP is very low)

Step 6. Oxidation of glyceraldehyde 3-phosphate

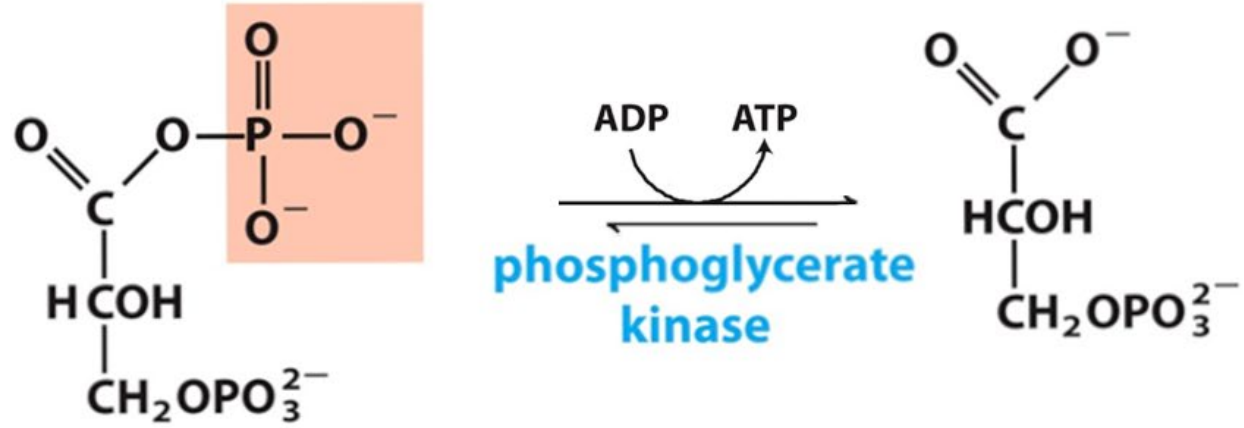


Lehninger 7th ed.

$\Delta G'° = 6.3$ kJ/mol

- **Aim**:
  - Generates a high-energy phosphate compound (acyl-phosphate group)
  - Gathers electrons (energy) as NADH (2 NADH molecules per molecule of glucose)
- Oxidation coupled to phosphorylation
- Reversible

Step 7. Substrate-level phosphorylation



1,3-Bisphosphoglycerate

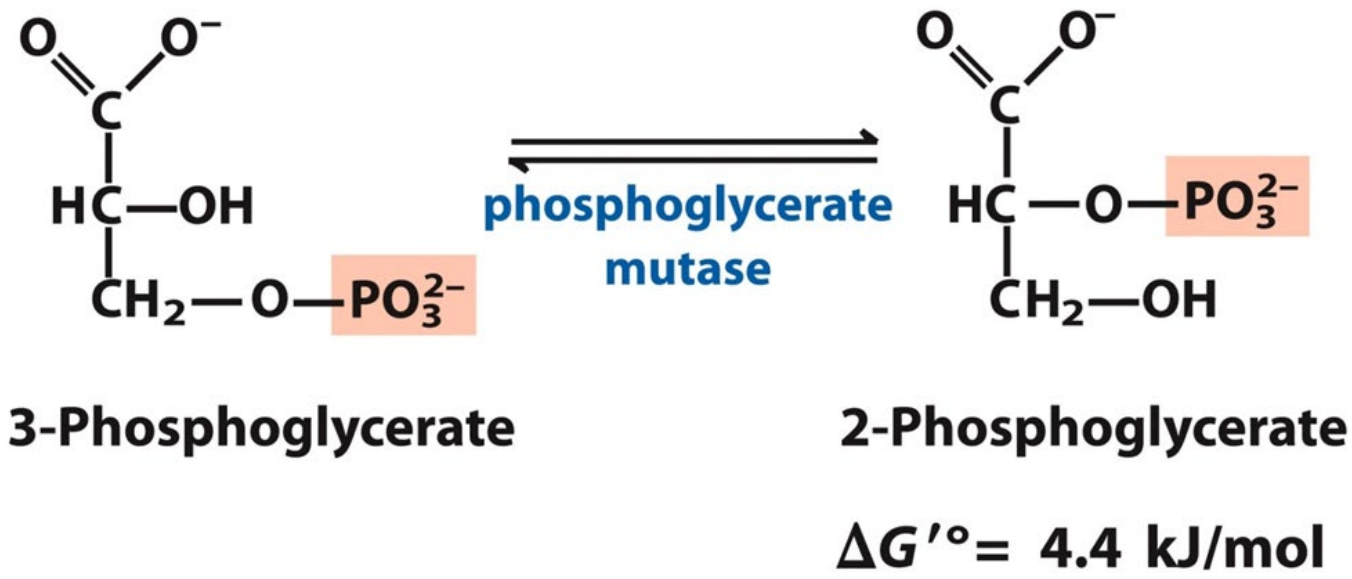
3-Phosphoglycerate

$\Delta G'° = -18.5$ kJ/mol

Lehninger 7th ed.

- **Aim**:
  - Makes ATP by substrate-level phosphorylation
- 2 ATP molecules per molecule of glucose
- Reversible

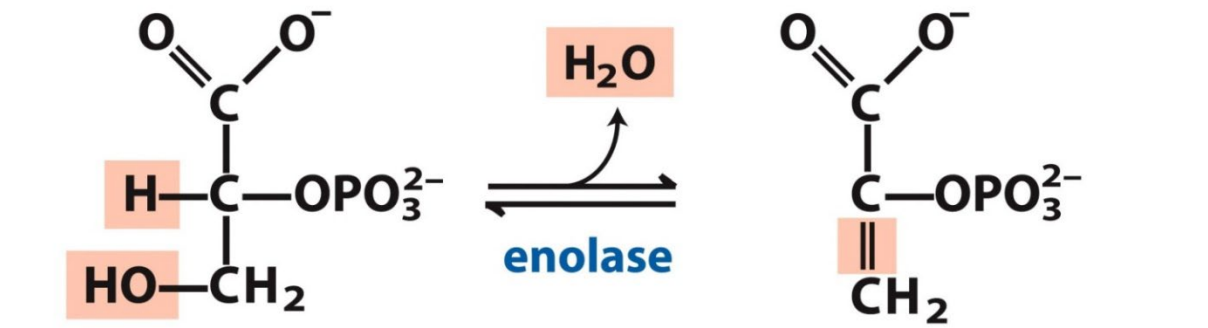
Step 8. Migration of the phosphate group



Lehninger 7th ed.

- **Aim**:
  - First step to form a high-energy phosphate compound
- Isomerization
- Reversible

Step 9. Dehydration of 2-phosphoglycerate



$\Delta G'° = 7.5$ kJ/mol

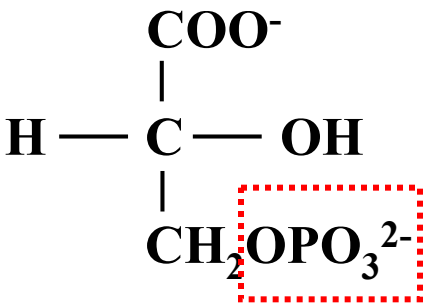
Lehninger 7th ed.

- **Aim**:
  - Generates a high-energy phosphate compound
- Reversible

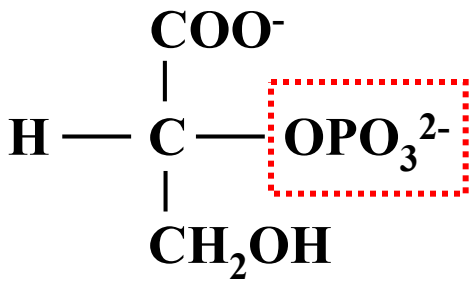
Phosphoryl-transfer potentials



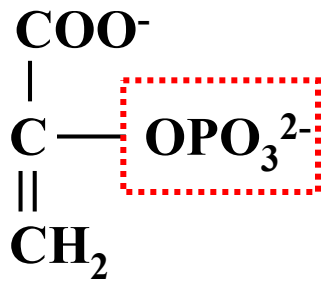
3-Phosphoglycerate



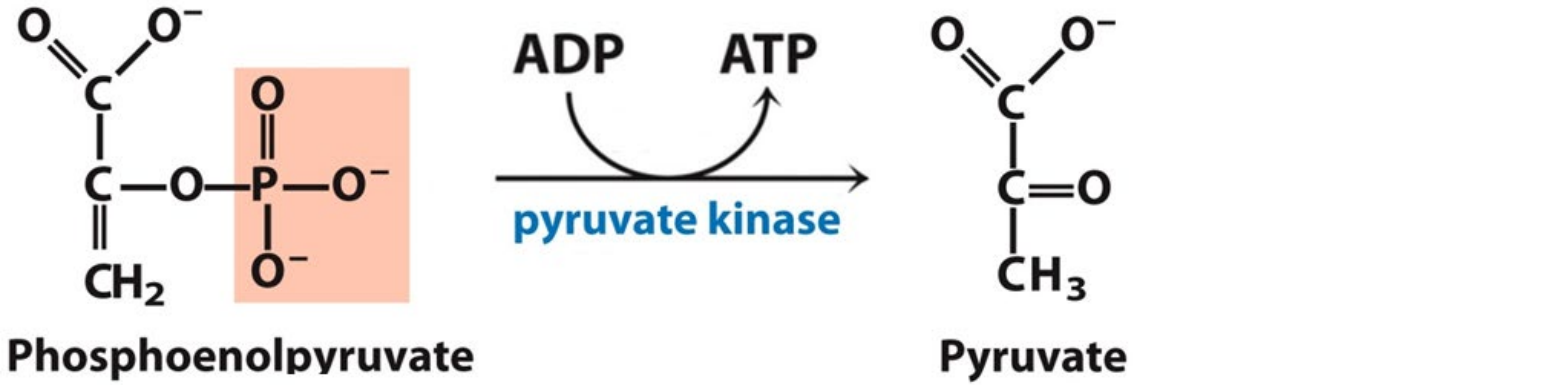
2-Phosphoglycerate



Phosphoenolpyruvate



Step 10. Substrate-level phosphorylation

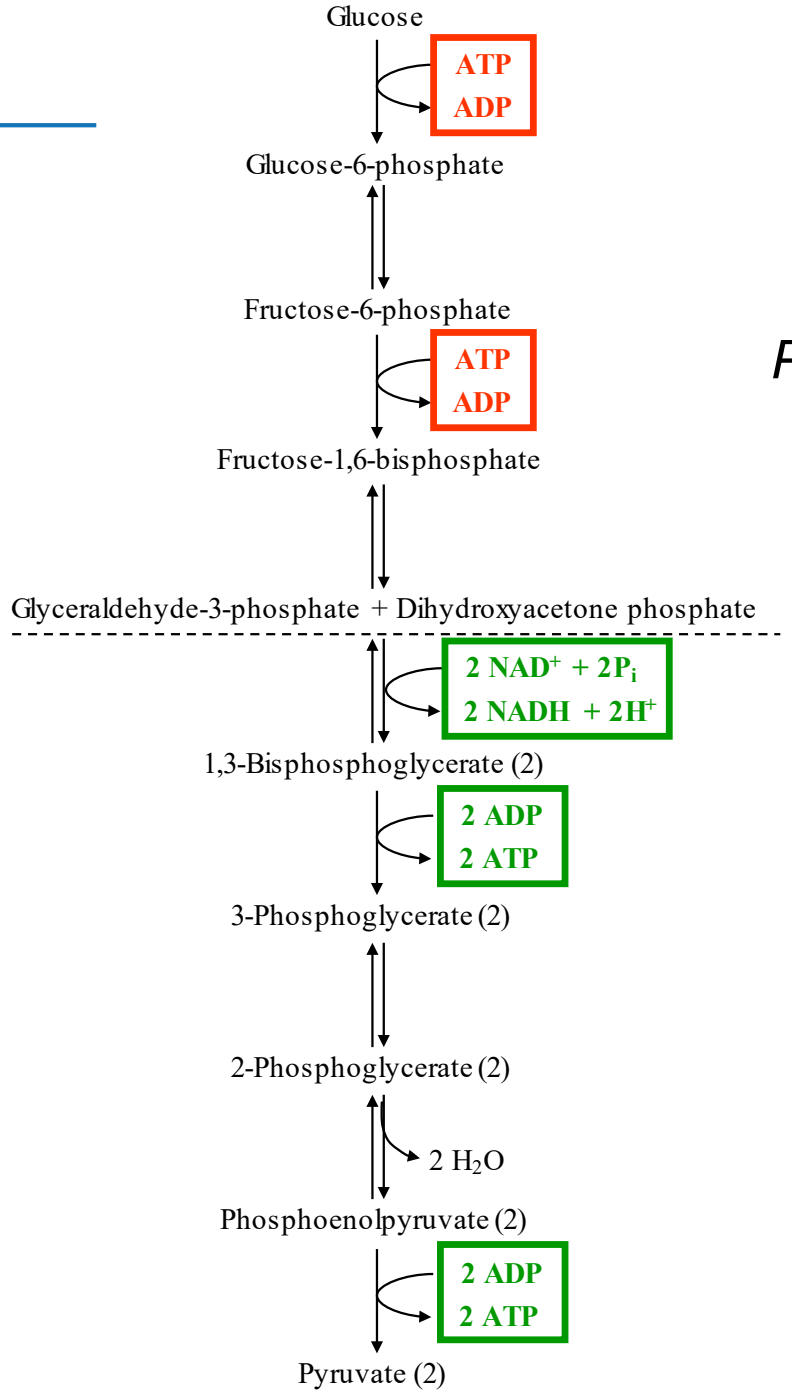


$\Delta G'° = -31.4$ kJ/mol

Lehninger 7th ed.

- **Aim**:
  - Generates ATP by substrate-level phosphorylation
- 2 ATP molecules per molecule of glucose
- Irreversible

Glycolysis overview



Preparatory phase

- 2 ATP

Payoff phase

+ 4 ATP

+ 2 NADH

Glycolysis overall balance

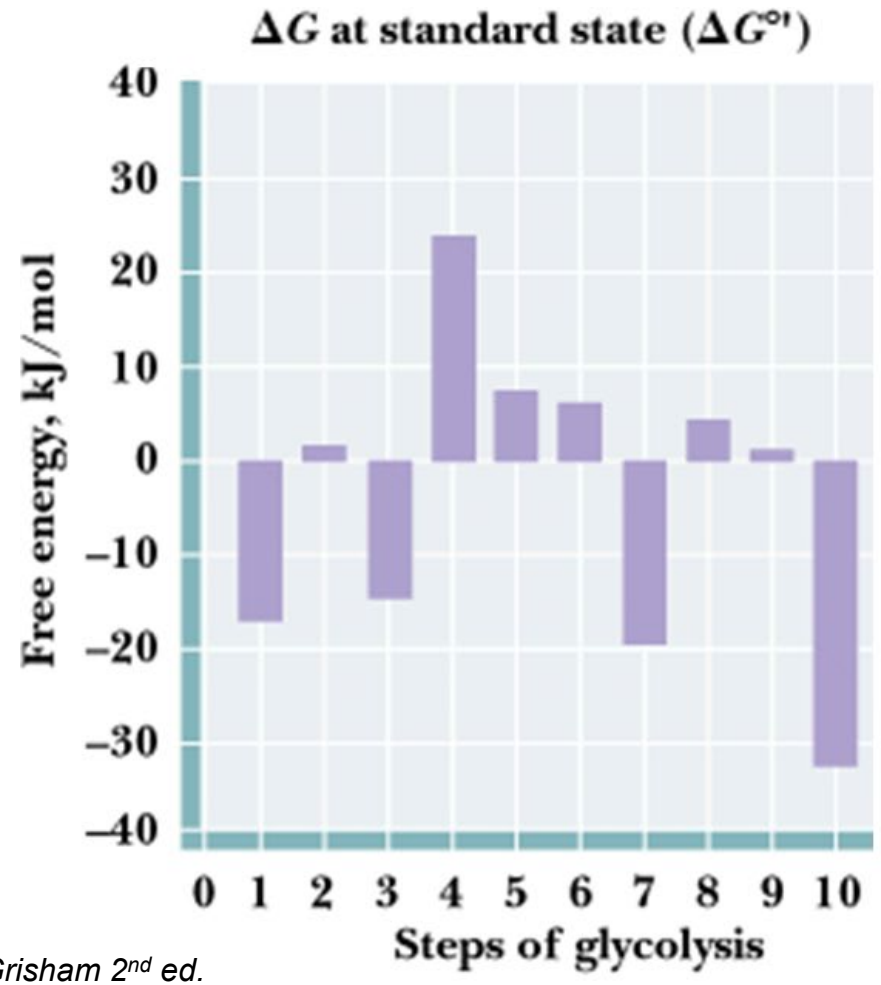
$$\text{Glucose} + 2\,\text{ADP} + 2\,P_i + 2\,NAD^+ \rightarrow$$

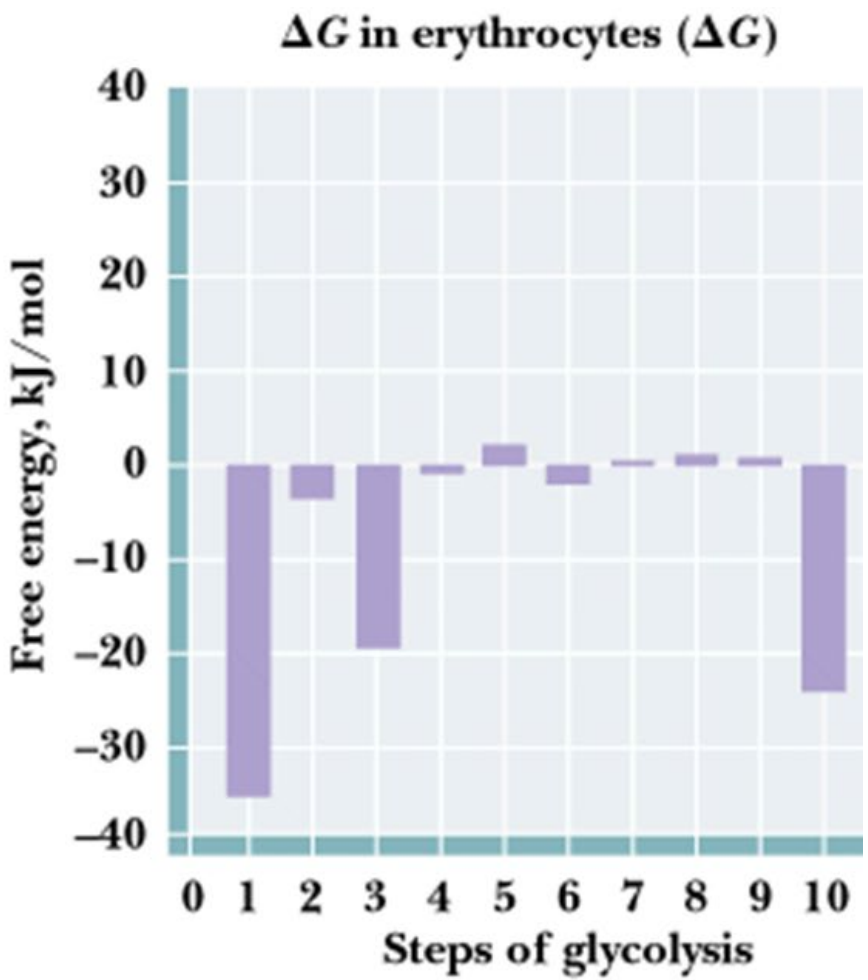
$$2\,\text{pyruvate} + 2\,\text{ATP} + 2\,\text{NADH} + 2\,H^+ + 2\,H_2O$$

Regulation of glycolysis

- Main purpose of glycolysis:
  - Production of energy (ATP)
  - Production of precursors for biosynthesis
- Key regulatory enzymes
  - Hexokinase
  - Phosphofructokinase-1
  - Pyruvate kinase

Thermodynamics of glycolysis





Garrett & Grisham 2nd ed.

Glycolysis is regulated at steps 1, 3 and 10

Regulation of hexokinase

Glucose + ATP → glucose 6-phosphate + ADP

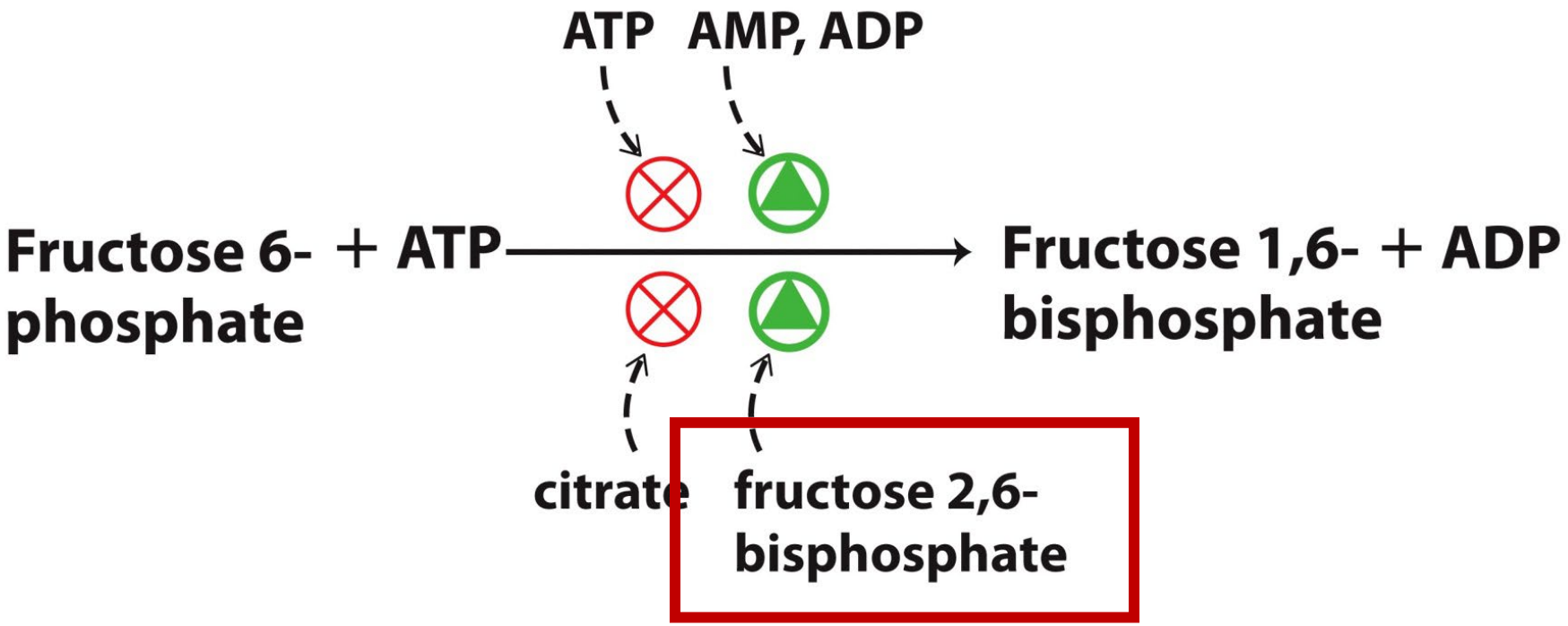
Hexokinase I

⊗ Glucose 6-phosphate

Hexokinase IV (Glucokinase)

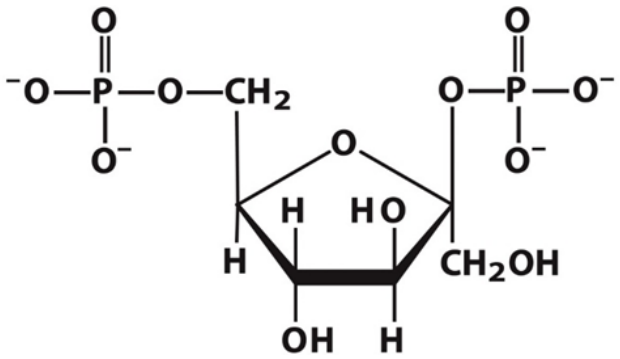
⊗ Fructose 6-phosphate

Regulation of phosphofructokinase-1



Lehninger 7th ed.

Production of fructose 2,6-bisphosphate



Allosteric activator of PFK1

NOT a glycolytic intermediate

Glucose

Glucose 6-P

Fructose 6-P

PFK-2

Fructose 2,6-BP

PFK-1

Fructose 1,6-BP

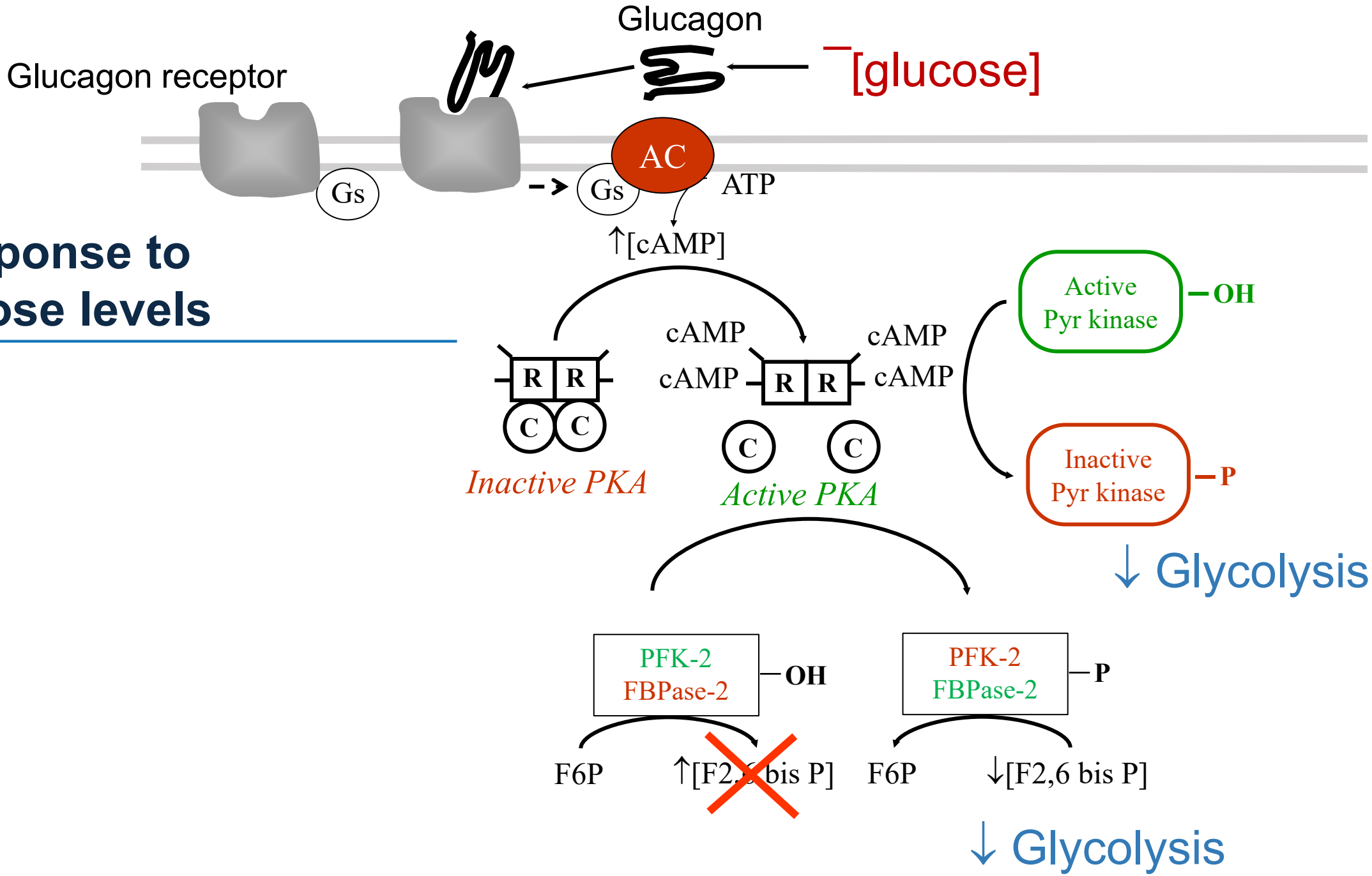
Pyruvate

Regulation of pyruvate kinase

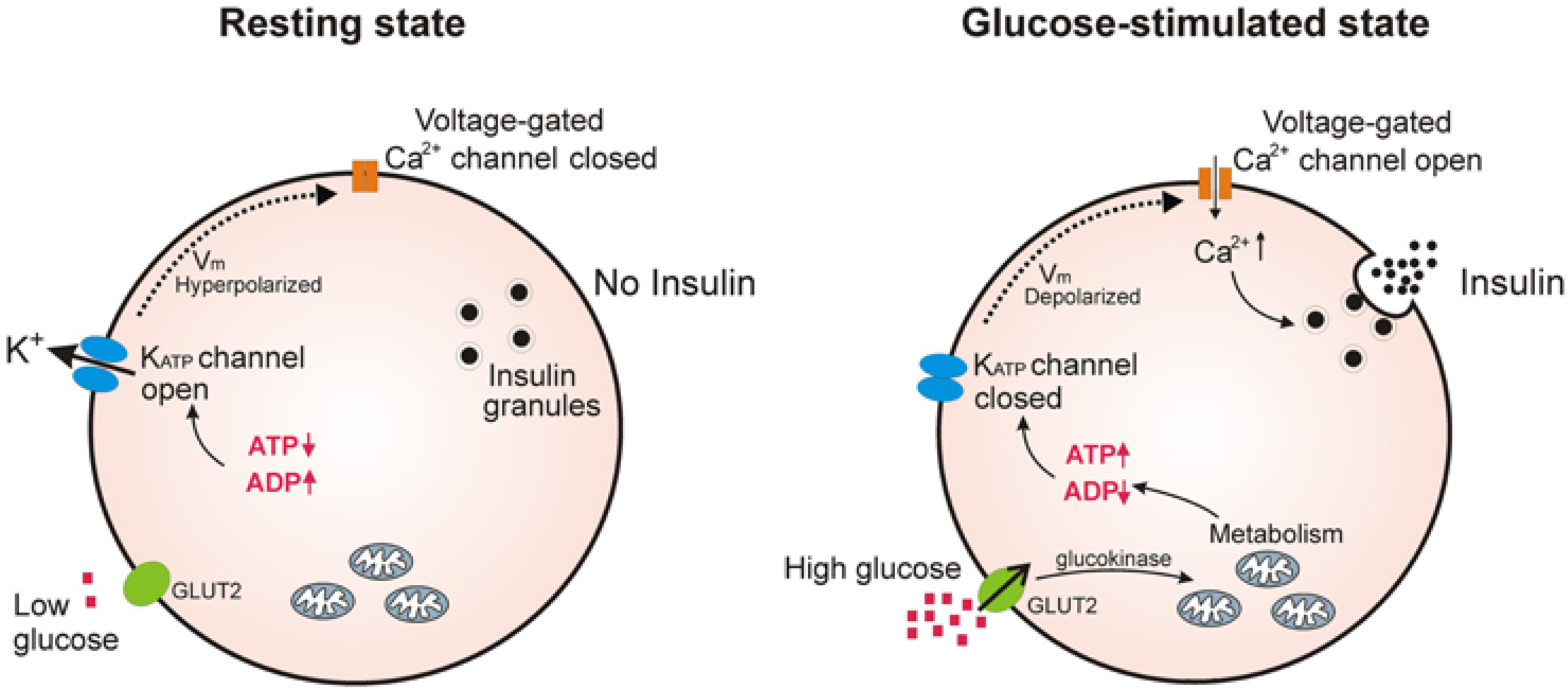
Phosphoenolpyruvate + ADP $\rightarrow$ pyruvate + ATP

- Allosterically activated by high flow through glycolysis, and allosterically inhibited by signs of abundant energy
- In the liver, inactivated by phosphorylation

Liver response to low glucose levels



Pancreas response to glucose levels



Martin M et al. Front. Physiol. 2013

Catabolic fates of pyruvate

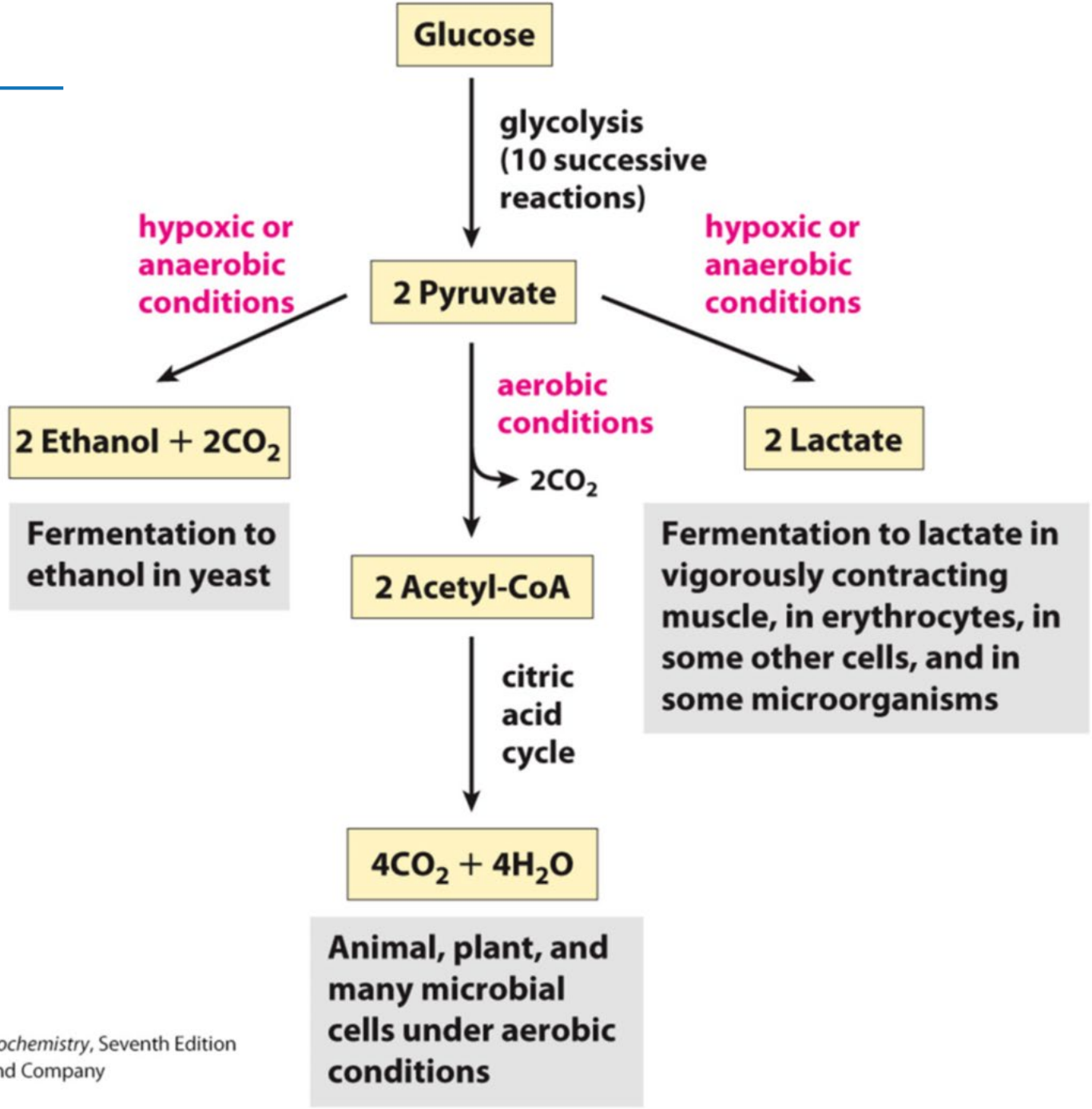
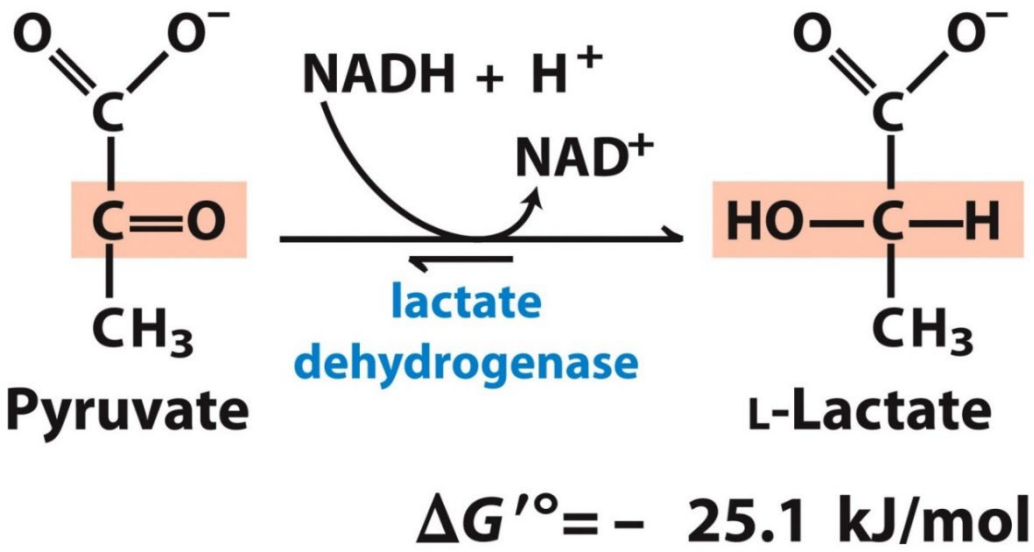


Figure 14-4
Lehninger Principles of Biochemistry, Seventh Edition
© 2017 W. H. Freeman and Company

Fermentation

- Glycolysis causes depletion of NAD^+ and accumulation of pyruvate in the cytosol
- The aim of fermentation is regenerate NAD^+ and eliminate pyruvate
- This allows for the generation of ATP from glucose without consuming oxygen
- In humans, this process is important in cells with impaired mitochondrial respiration (e.g., erythrocytes) and during high-intensity exercise (when there is shortage of O_2 supply)

Lactic fermentation

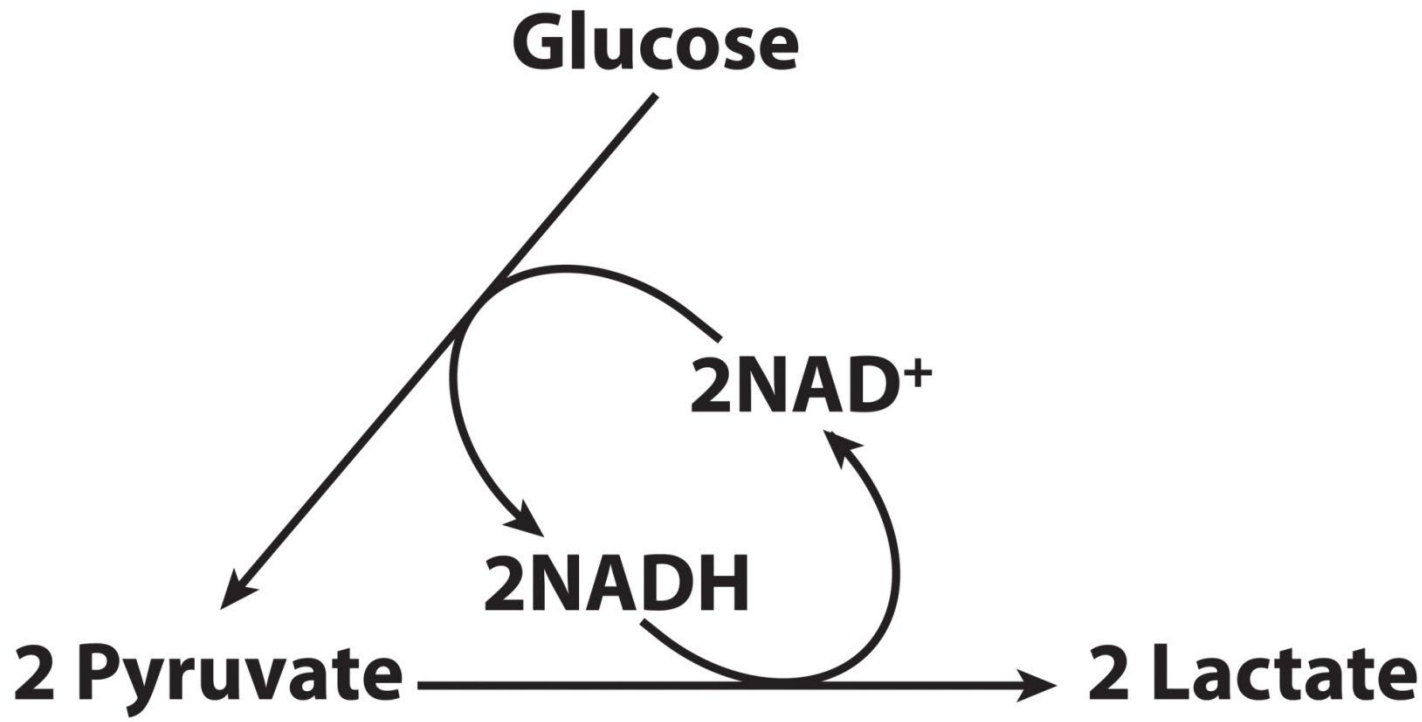


Lehninger 7th ed.

- Reversible reduction of pyruvate to lactate
- During strenuous exercise, lactate builds up in the skeletal muscle
- The acidification of muscle during this process prevents a continuous strenuous work

Lactic fermentation: Balance

$$\text{Glucose} + 2\ \text{ADP} + 2\ P_i \rightarrow 2\ \text{lactate} + 2\ \text{ATP} + 2\ H_2O$$



Lehninger 7th ed.

The Warburg effect

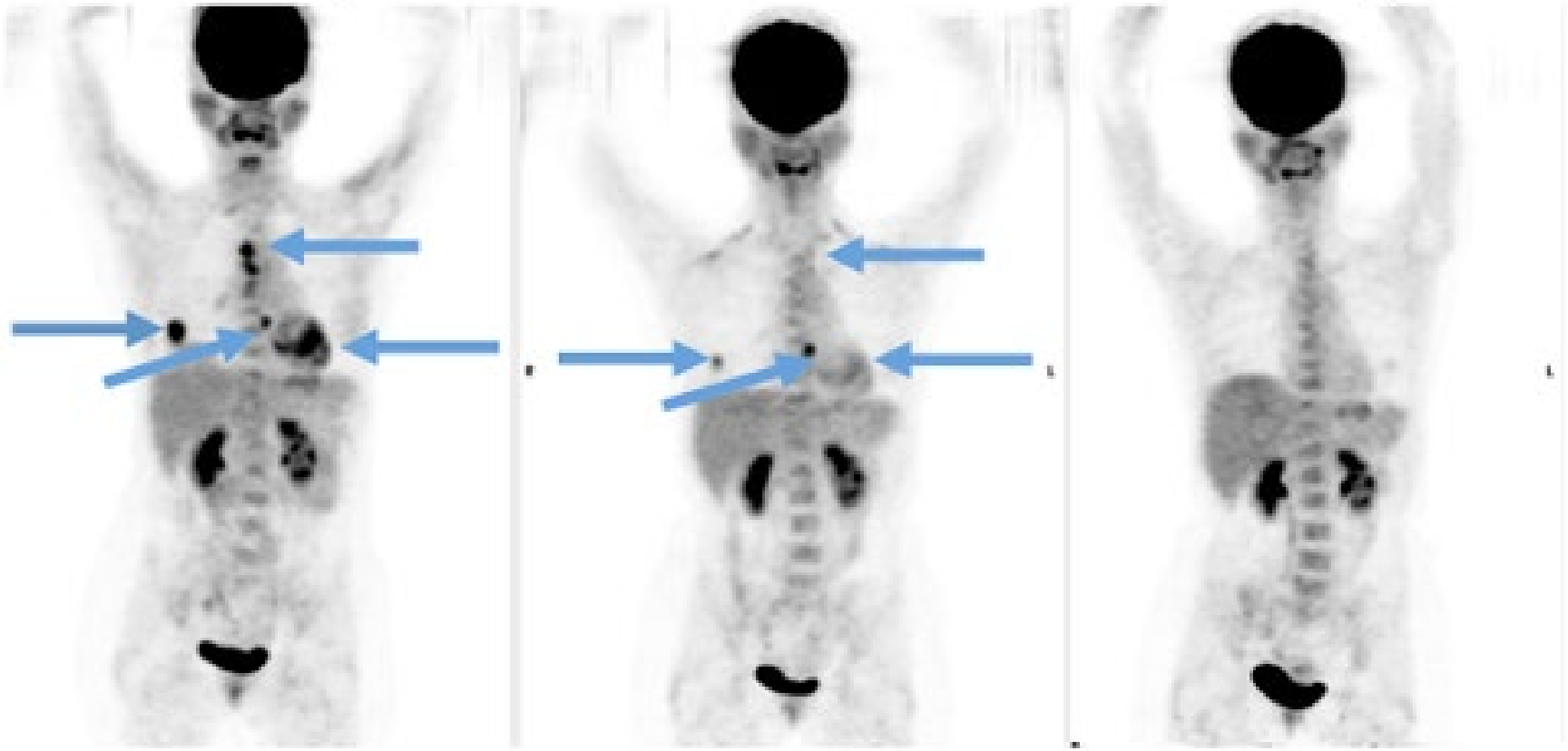
"Cancer, above all other diseases, has countless secondary causes. But, even for cancer, there is only one prime cause. Summarized in a few words, the prime cause of cancer is the replacement of the respiration of oxygen in normal body cells by a fermentation of sugar."
-- Dr. Otto H. Warburg (1924)



Tumors have a high glucose utilization rate

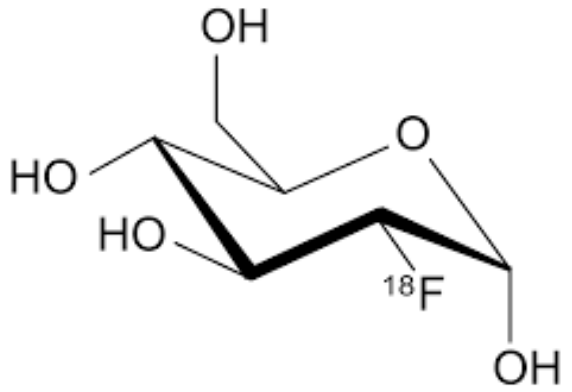
^{18}Fluorodeoxyglucose PET image

Treatment response with PET FDG in non-small cell lung cancer



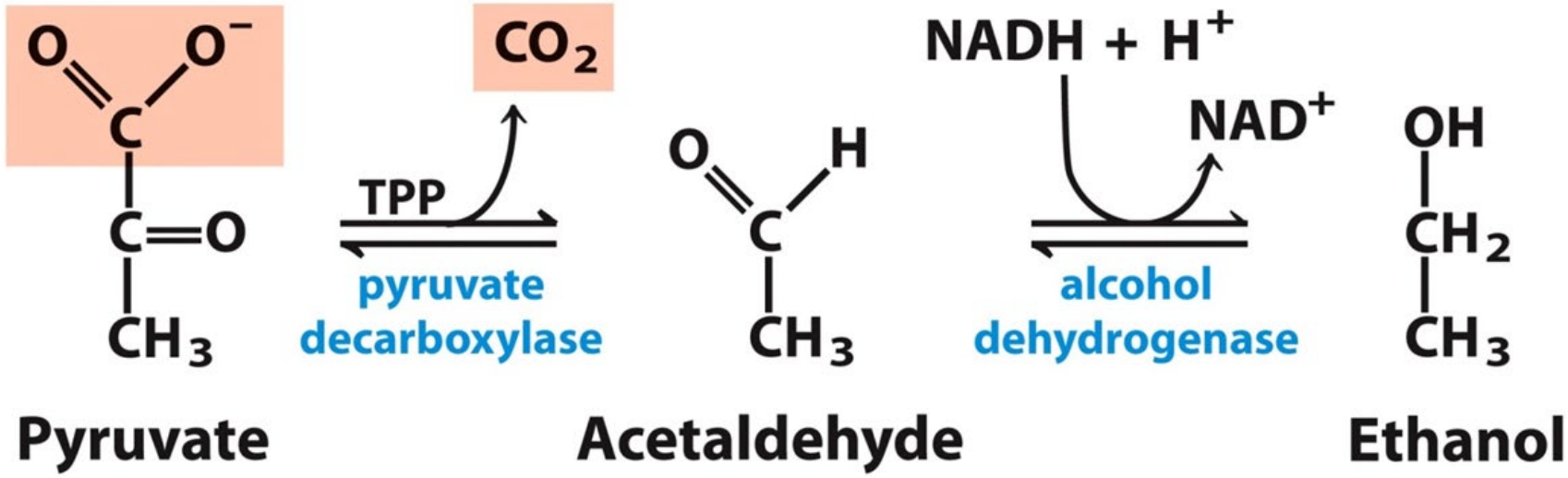
Baseline 2 weeks 6 weeks

Time after start of treatment



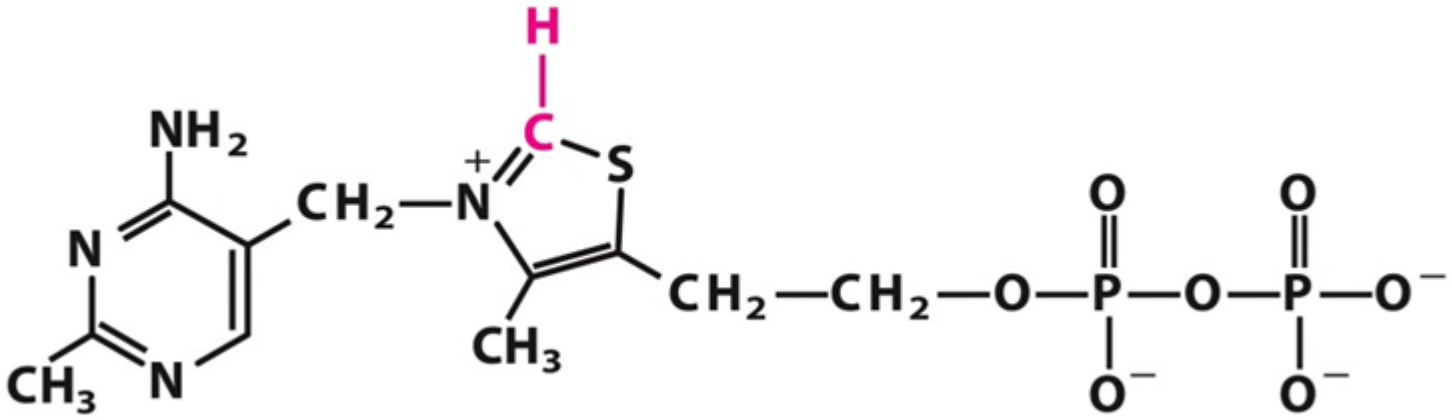


Ethanol fermentation



- Two-step reduction of pyruvate to ethanol
- Humans do not have pyruvate decarboxylase, but we do express alcohol dehydrogenase for ethanol metabolism

Thiamine pyrophosphate



- It is a thiamine (vitamin B_1) derivative
- It is a cofactor that is present in all living systems, in which it catalyzes several biochemical reactions

Entry of other carbohydrates into glycolysis

